

Algorithm Formula

Score formula is:

$$SC = \frac{(clicks \cdot w_c + votes \cdot w_v + comments \cdot w_m)}{(currentTime - postTime + 2)^G}$$

- ➔ Clicks, votes, comments get a certain weight (0.3, 0.4, 0.3)
- ➔ currentTime, postTime both in minutes
- ➔ +2 prevents division by zero and avoids spikes. If we remove +2, the score goes to infinity when the post is new.

Simpler:

$$score = \frac{E}{(t + C)^G}$$

- ➔ E is event
- ➔ t is time
- ➔ C is so called constant offset
- ➔ G is gravity

Gravity

If G is **small** (e.g. 0.5), older posts lose score slowly.

If G is **large** (e.g. 1.5), older posts lose score quickly.

e.g.

Reddit uses something close to **1.8**.

News or fast-paced apps might use **1.5–2.0**.

Slower content (like blogs) might use **0.8–1.2**.

In our case, we'll use 1.2

Weights

We want to push posts which are engaged with:

1. Number of votes
2. Number of comments
3. Number of clicks

Votes: 0.6

Comments: 0.3

Clicks: 0.1

But:

Post X: 100 clicks, 20 votes, 5 comments => 23.5

Post Y: 300 clicks, 5 votes, 2 comments => 33.6

Which states that the clicks are more important than the votes. Big L. Not what we want.

So, to put everything to the same scale, normalization:

$$\text{normalized_clicks} = \frac{\text{clicks}}{\text{avg_clicks_per_post}}$$

$$\text{normalized_votes} = \frac{\text{votes}}{\text{avg_votes_per_post}}$$

$$\text{normalized_comments} = \frac{\text{comments}}{\text{avg_comments_per_post}}$$

So, the full finalized formula:

$$SC = \frac{(\text{normalized_clicks} \cdot 0.6 + \text{normalized_otes} \cdot 0.3 + \text{normalized_comments} \cdot 0.1)}{(\text{currentTime} - \text{postTime} + 2)^G}$$

Optional Comments Weights

$$\text{Comment Score} = \text{base} + b_l + b_r$$

e.g.

$$b_l = \text{length_bonus} = 0.6 * ((\text{length_chars} \% 100) / 100)$$

$$b_r = \text{replies_bonus} = 0.12 \times \min(\text{numberReplies}, 10)$$

➔ 10 is br is the max_replies_cap

Potential Considerations

➔ Posts which are on 50% - 50% level for some time would get pushed to trending